



RESEARCH SOFTWARE / METHODS AND VALIDATION STATUS / VERSION 0.3

# NeuroCause: an inspectable platform for connectome-constrained fly simulation

Architecture, data provenance, internal verification, negative results and prospective validation criteria

NeuroCause Research Software | [contact@neuro-cause.com](mailto:contact@neuro-cause.com)

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## ABSTRACT

**Background.** Synapse-resolution connectomes constrain neural connectivity but do not determine physiology, behavior, drug response or subjective experience. **Methods.** NeuroCause couples a selected adult-female FAFB/FlyWire brain graph to a selected adult-male MaleCNS nerve-cord graph and virtual body. The source animals meet only through an explicit modelled population-rate bridge. Female BANC and male MANC explorers remain separate; an optic-lobe explorer uses an earlier overlapping release of the MaleCNS specimen.

**Implementation.** The running brain contains 7,270 neurons and 784,219 directed connection rows; the cord contains 1,045 neurons and 17,224 rows. Versioned source identities and seeded tests support internal verification, not held-out animal validation. **Interpretation.** This is an intervention-capable research prototype, not a sex-consistent digital animal, a validated pharmacology assay or evidence of digital sentience. We state the empirical gates required for those aims.

**Keywords:** connectomics; *Drosophila melanogaster*; embodied simulation; provenance; model validation; sentience

## 1 Introduction

Adult *Drosophila* connectomics now provides brain-scale and brain-and-cord reconstructions from identified female and male specimens [1-4]. Anatomical cell identities and putative contacts constrain possible information flow, but do not specify receptor expression, synaptic gain, ion-channel kinetics, sensory transduction or muscle dynamics [5]. A connectome-derived simulation is consequently a set of testable assumptions rather than a replay of the source animal.

Published connectome-constrained models have produced experimentally testable sensorimotor predictions [6,7], while neuromechanical simulators show the importance of explicit body-environment coupling [8]. NeuroCause aims to combine these lines in a reproducible in-silico laboratory. This methods paper documents the present implementation, its missing measurements and negative results, then defines prospective criteria for stronger biological claims.

## 2 Data provenance and specimen identity

The current anatomy explorer contains bounded, independently identified subgraphs; it does not contain complete published CNS graphs. Edge rows are directed neuron pairs with anatomical contact counts. A count is neither a measured conductance nor a confirmed excitatory or inhibitory postsynaptic effect [5]. Table 1 is counted from the versioned local catalog; the public v2.2.0 source snapshot [S1] does not yet include every exploratory bundle.

| Source / animal                   | Project anatomy subset                         | Runtime relationship                                      |
|-----------------------------------|------------------------------------------------|-----------------------------------------------------------|
| FAFB / FlyWire v783; female brain | 6,000 cells; 268,248 pairs; 6,116,021 contacts | Explorer; expanded FAFB selection drives the active brain |
| BANC v888; female brain + cord    | 6,000 cells; 167,668 pairs; 3,634,624 contacts | Explorer only; candidate for a future same-animal runtime |
| MaleCNS v1.0; male brain + cord   | 6,000 cells; 305,504 pairs; 8,402,924 contacts | Explorer; separate cord selection in active mixed runtime |
| MANC v1.0; male nerve cord        | 6,000 cells; 316,619 pairs; 7,655,223 contacts | Explorer only; a different individual from MaleCNS        |
| Optic lobe v1.1; male right lobe  | 6,000 cells; 149,806 pairs; 2,443,048 contacts | Explorer only; same MaleCNS sample, older release         |

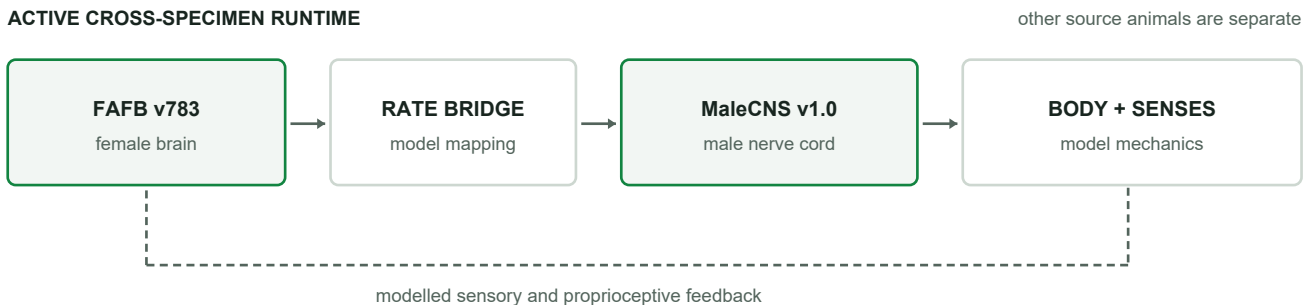
**Table 1.** Each explorer bundle is capped at 6,000 selected cells and retains induced pairs with at least five contacts. The bundles are not complete CNS reconstructions and are not interchangeable animals.

A dated raw-source snapshot records 27 checksum-matching files (3.19 GB) across BANC, MaleCNS, MANC v1.0, hemibrain v1.2 and male optic-lobe v1.1. These digests were compared with earlier *local* SHA-256 records, not independently authenticated publisher checksums. Three checked MANC files and two optic-lobe files support bounded anatomy explorers; hemibrain remains raw-only. The optic-lobe sample is the same biological individual as MaleCNS, while MANC and hemibrain are separate animals [9–11].

The optic-lobe v1.1 explorer selects one located traced cell per named type, then fills to 6,000 by source synaptic-weight ranking; 6,000/6,000 selected body IDs are present in local MaleCNS v1.0 annotations. This establishes ID overlap for the selection, not equivalence of versioned segmentation or all edges. Its directed pairs are never summed with the newer MaleCNS graph. The 8-nm EM voxel scale is used for displayed soma coordinates [11].

### 3 Simulation architecture

#### ACTIVE CROSS-SPECIMEN RUNTIME



**Figure 1.** Outlined graph blocks contain published anatomical connectivity; their spiking dynamics, the cross-specimen bridge, body, sensory encoders and feedback are model-defined. BANC and archived reference animals are not combined into this circuit.

#### 3.1 Neural state and closed loop

The active FAFB-derived brain has a 6,338-cell core (703,381 directed rows), a 68-cell temperature extension (4,749 rows), and an 864-cell taste and antennal-grooming extension (76,089 rows): 7,270 neurons and 784,219 rows in total. The MaleCNS-derived locomotor selection has 1,045 cells, 17,224 directed rows and 708,689 selected contacts. These rows are selected graph edges, not all published synapses. Membrane constants, delay, baseline drive, noise and efficacy are model parameters; predicted transmitter classes do not settle receptor effects [5].

A 120-Hz world/body loop advances finer neural steps under a deterministic seed. The female-brain to male-cord interface transfers same-type/side population rates: it is an engineered homologous mapping, not observed cross-animal synapses. Display exposure and observer overlays are isolated from the fly-eye rendering path. Leg stepping, proprioception, motor force and contact are model rules whose calibration is incomplete.

### 3.2 Stimuli, interventions and exports

Model encoders address visual, antennal, gustatory and temperature populations. Olfactory receptor neurons are absent from the running subset, so an environmental odor field is not a validated neural input. Users can stimulate, silence or activate selected populations, alter generic transmitter gains and export traces with seeds, source fingerprints and intervention manifests. A within-model perturbation establishes dependence on the implemented graph and rules, not equivalence with an intact fly.

### 3.3 Evidence classes and observables

The application separates source-derived observations from latent quantities introduced by the simulator. A provenance label is not a confidence interval; uncertainty must be estimated per measurement and per fitted parameter before quantitative biological use.

| Class                 | Directly available                                                    | Not established by this class                             |
|-----------------------|-----------------------------------------------------------------------|-----------------------------------------------------------|
| Reconstructed anatomy | Versioned cell identities, selected directed pairs and contact counts | Conductance, receptor effects, complete animal physiology |
| Simulated dynamics    | Seeded membrane states, spikes, behavior and intervention traces      | Match to withheld recordings or another animal            |
| Subjective experience | No direct observable in the current software                          | Feeling, pain or sentience in the model                   |

**Table 2.** Evidential layers. A computed state may be useful as a hypothesis even when it has not been empirically calibrated.

## 4 Internal verification and negative evidence

In the development version inspected on 28 September 2026, **npm test** passed locally on Windows with Node.js 24.19.0; **npm run specimens:uitest** passed in a headless Edge browser for all five anatomy profiles. The checks exercise specimen identity, graph validity, seeded reproducibility, sensory routing, closed-loop mechanics, exports and scientific claim boundaries. They are software-invariant checks. No held-out neural recording, across-animal generalization test, blinded compound-response benchmark or independent replication is included here.

| Question             | Developer-documented within-model result                                                               | Interpretation                                       |
|----------------------|--------------------------------------------------------------------------------------------------------|------------------------------------------------------|
| Looming escape       | Abrupt loom activates the giant-fiber path; visual-route silencing suppresses it.                      | Internal mechanism test; in-vivo calibration pending |
| Taste / grooming     | Sugar drives proboscis output; bitter suppresses it. JO-F and DNg12 perturbations alter dust grooming. | Within-model causal dependence                       |
| Habituation          | Repeated stimulation does not show the expected decline in the documented protocol.                    | Not reproduced                                       |
| Associative learning | Paired versus reversed result was not significant (documented $p = 0.19$ ).                            | No demonstrated learning effect                      |
| Thermal preference   | Documented occupancy near 25 C was approximately 15% versus 16%.                                       | Not reproduced                                       |

**Table 3.** Selected internal results from the versioned validation log [S1]. Negative outcomes remain visible. These observations are not prospective animal-level accuracy or evidence of experience.

Verification and validation are distinct. Software tests can show that a stimulus reaches an annotated population and that a model output changes reproducibly; they do not establish that gain, timing or behavioral distributions match a living animal. Biological validation requires frozen parameters, withheld data, prespecified failure criteria, uncertainty intervals and comparison with simpler baselines.

## 5 Limits on interpretation

### 5.1 Anatomy and physiology

Segmentation, synapse detection and annotation contain uncertainty; bounded subgraphs may remove weak but decisive paths. A predicted transmitter does not measure receptor expression or synaptic efficacy [5]. The virtual eye, muscles, legs and environment were not measured from the reconstructed individual. Shared cell names or matching sex never justify synthetic cross-specimen synapses. The current mixed-sex runtime is a transparent engineering test bed, not a one-animal reconstruction.

### 5.2 Pharmacology

Predicting a named compound requires molecular targets, cell-specific expression, concentration-response curves, uptake and metabolism, time course and blinded dose-response validation. Generic gain or blockade controls are circuit interventions, not drug predictions. NeuroCause cannot currently replace live-fly assays, preclinical efficacy tests or regulatory toxicology [12].

### 5.3 Sentience

Research on living insects provides substantive yet debated evidence concerning pain-related processing and behavior [13-15]. Such evidence does not transfer automatically to software. Theory-derived indicators of AI consciousness guide hypotheses but are not a validated meter of felt states [16]. NeuroCause reports modeled variables, not measured feelings, pain or a sentience score. Digital sentience remains an explicit long-term research question requiring independent conceptual, computational and ethical assessment; this implementation offers no demonstration of it.

## 6 Prospective research program

Four gates should be assessed independently. **Specimen coherence:** construct a native, non-duplicated brain-to-cord graph from one BANC female or one MaleCNS male and audit coverage and missing edges. **Physiological calibration:** fit identifiable cell-class, sensory and motor parameters to independent recordings and perturbations, with uncertainty. **Prospective validation:** preregister endpoints, freeze code and parameters, and predict withheld animals, genotypes, stimuli and eventually compounds against live data and simpler baselines. **Sentience assessment:** define competing theories and discriminating tests before welfare-relevant attribution, with external review and precaution under uncertainty.

The near-term scientific question is whether a specified model predicts withheld neural and behavioral data better than simpler alternatives within a declared domain. Even strong prediction alone would not prove animal equivalence, clinical utility or subjective experience.

## 7 Transparency and availability

**Study type:** public, non-peer-reviewed research-software working paper; no new animal experiments. **Source:** NeuroCause application v2.2.0 and a development anatomy explorer, inspected 28 September 2026. The public software snapshot [S1] covers the released runtime but does not yet contain the MANC and optic-lobe exploratory bundles reported here. Publisher datasets are linked below. Exact reproduction of those new explorer counts requires a later public importer and generated-asset release; independent validation additionally requires frozen parameters and benchmark outputs.

**Licensing and interests:** FAFB-derived data have CC BY-NC 4.0 conditions. BANC, MaleCNS, MANC and optic-lobe source terms must be respected separately. NeuroCause develops the software discussed here. No external validation, clinical use or commercial drug-screening permission is implied.

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